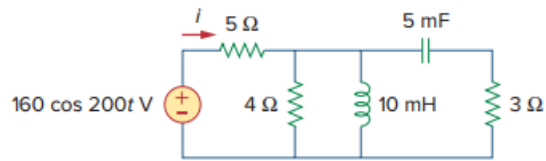


Problem

Calculate $i(t)$ in the circuit of Fig. 9.51.

Figure 9.51:



Step-by-step solution

Step 1 of 3

The voltage source is,

$$v(t) = 160 \cos(200t) \text{ V}$$

The angular velocity ω is 200 rad/s.

Convert the voltage into phasor domain.

$$\mathbf{V} = 160 \angle 0^\circ \text{ V}$$

Determine the capacitive impedance, $\mathbf{Z}_C$.

$$\begin{aligned} \mathbf{Z}_C &= \frac{1}{j\omega C} \\ &= \frac{1}{j(200)(5 \times 10^{-3})} \\ &= -j \Omega \end{aligned}$$

Determine the inductive impedance, $\mathbf{Z}_L$.

$$\begin{aligned} \mathbf{Z}_L &= j\omega L \\ &= j(200)(10 \times 10^{-3}) \\ &= j2 \Omega \end{aligned}$$

Step 2 of 3

Determine the equivalent impedance $\mathbf{Z}_{eq}$ of the network.

$$\begin{aligned} \mathbf{Z}_{eq} &= 5 + [4 \parallel \mathbf{Z}_L \parallel (3 + \mathbf{Z}_C)] \\ &= 5 + [4 \parallel (j2) \parallel (3 - j)] \\ &= 5 + \left[4 \parallel \frac{(j2)(3 - j)}{j2 + 3 - j} \right] \\ &= 5 + [4 \parallel (1.2 + j1.6)] \\ &= 5 + \frac{(4)(1.2 + j1.6)}{4 + 1.2 + j1.6} \\ &= 6.189 + j0.86486 \Omega \end{aligned}$$

Step 3 of 3

Determine the current from the source.

$$\begin{aligned}\mathbf{I} &= \frac{\mathbf{V}}{\mathbf{Z}_{\text{eq}}} \\ &= \frac{160\angle 0^\circ}{6.189 + j0.86486} \\ &= 25.357 - j3.543 \text{ A} \\ &= 25.6\angle -7.96^\circ \text{ A}\end{aligned}$$

Express the current in time domain.

$$i(t) = 25.6\cos(200t - 7.96^\circ) \text{ A}$$

Thus, the current $i(t)$ in the circuit is $25.6\cos(200t - 7.96^\circ) \text{ A}$.